# An Exact Anosov Orbit Zeta and a Koopman Determinant-Class Obstruction

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**Abstract**

For the toral automorphism induced by $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, we prove $\#\,\mathrm{Fix}(T_A^n) = |\det(A^n - I)|$ at every period, recover all primitive counts by Möbius inversion, and resum the Artin–Mazur zeta to $(1-z)^2/(1-3z+z^2)$. The natural Koopman action on $L^2(\mathbb{T}^2)$ is unitary but noncompact: it permutes the Fourier basis and therefore belongs to no finite Schatten class. Thus the exact orbit zeta is not the ordinary trace-class Fredholm determinant of this natural owner. Parabolic, signed-count, and cyclic-aliasing controls delimit the result.

## 1 Complete orbit census

Freeze $T_A([x]) = [Ax]$ on $\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$, with one iterate as clock and unsigned fixed-point cardinality as normalization. The eigenvalues of $A$ are

$$\lambda_\pm = \frac{3 \pm \sqrt{5}}{2}, \qquad \lambda_+ > 1 > \lambda_- > 0, \qquad \lambda_+\lambda_- = 1.$$

Hence $A^n - I$ is nonsingular for every $n \geq 1$. Fixed points of $T_A^n$ are the kernel of the induced integer torus endomorphism, whose cardinality is the lattice index $|\det(A^n - I)|$.

**Proposition 1.** *Let $S_n = \mathrm{tr}(A^n)$. For every $n \geq 1$,*

$$N_n := \#\,\mathrm{Fix}(T_A^n) = S_n - 2, \quad S_0 = 2,\ S_1 = 3,\ S_n = 3S_{n-1} - S_{n-2}.$$

*The number of points of exact period $n$ is $P_n = \sum_{d|n} \mu(n/d) N_d$, and the primitive-orbit count is $O_n = P_n/n$.*

Indeed, in dimension two, $\det(A^n - I) = \det(A^n) - \mathrm{tr}(A^n) + 1 = 2 - S_n < 0$. Möbius inversion then separates primitive points from repetitions. This is an all-order theorem; Table 1 is only a replay sentinel.

Table 1: Exact replay through period twelve.

| $n$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $N_n$ | 1 | 5 | 16 | 45 | 121 | 320 |
| $O_n$ | 1 | 2 | 5 | 10 | 24 | 50 |
| $n$ | 7 | 8 | 9 | 10 | 11 | 12 |
| $N_n$ | 841 | 2205 | 5776 | 15125 | 39601 | 103680 |
| $O_n$ | 120 | 270 | 640 | 1500 | 3600 | 8610 |

## 2 Primitive product and exact zeta

With the frozen unweighted convention,

$$\zeta_T(z) = \exp\left(\sum_{n\geq 1} \frac{N_n z^n}{n}\right) = \prod_{\gamma \text{ primitive}} (1 - z^{|\gamma|})^{-1}.$$

Since $N_n = \lambda_+^n + \lambda_-^n - 2$, the elementary logarithmic series gives

$$\boxed{\zeta_T(z) = \frac{(1-z)^2}{(1-\lambda_+ z)(1-\lambda_- z)} = \frac{(1-z)^2}{1-3z+z^2}.}$$

The identity holds formally at zero and as the displayed rational meromorphic function. It is an internally owned orbit zeta, not a target-divisor match.

# 3 Natural Koopman obstruction

On $\mathcal{H} = L^2(\mathbb{T}^2, dx)$ define $(Uf)(x) = f(T_A x)$. Haar invariance makes $U$ unitary. For the Fourier characters $e_k(x) = \exp(2\pi i k \cdot x)$,

$$Ue_k = e_{A^{\mathsf{T}}k}.$$

The index map is a bijection of $\mathbb{Z}^2$. More decisively, the orthonormal sequence $e_{(j,0)}$ maps to the orthonormal sequence $e_{(2j,j)}$. Its image has no norm-convergent subsequence, so $U$ is not compact. Consequently $U$ lies in no finite Schatten class, is not trace class, and supplies neither an ordinary operator trace nor an ordinary trace-class Fredholm determinant $\det(I - zU)$. We therefore do not identify that nonexistent ordinary determinant with $\zeta_T$; generalized flat or anisotropic traces would require a separately frozen theorem.

# 4 Three exact controls

First, the parabolic parent $B = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ satisfies $B^n = \left(\begin{smallmatrix} 1 & n \\ 0 & 1 \end{smallmatrix}\right)$. Its fixed set is the union of the $n$ circles $y = j/n$; the finite fixed-cardinality convention is unavailable. Second, $\det(A^n - I) = -N_n$: deleting the absolute value creates signed Lefschetz data, not negative cardinalities.

Third, wrap Fourier indices modulo $m$ and count solutions of $(A^{\mathsf{T}n} - I)k = 0 \pmod m$. These finite cyclic pseudo-traces vary with $m$ rather than approaching an ordinary Koopman trace. For example,

| $n\backslash m$ | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| 2 | 1 | 1 | 1 | 5 | 1 | 1 | 1 |
| 3 | 4 | 1 | 16 | 1 | 4 | 1 | 16 |
| 4 | 1 | 9 | 1 | 5 | 9 | 1 | 1 |
| 6 | 4 | 1 | 16 | 5 | 4 | 1 | 64 |

The evidence exhausts moduli 2 through 12 for these four iterates.

# 5 Progress, validation, and boundary

Relative to C121, the result advances from all-order degree growth plus one cycle to a complete all-order fixed/primitive census and exact orbit zeta. Relative to C119, rich recurrent dynamics and a natural global Hilbert action now coexist in one system; the new theorem identifies their precise mismatch: the natural action is unitary rather than determinant class.

An independent implementation reconstructs every evidence field; a fresh SymPy program passes 238 exact checks, canonical replay fixes the bytes, and all 23 hostile edits are rejected. Fixed-date isolated PDF builds agree bytewise and all fonts are embedded.

Complete intrinsic orbit data still lack a prime-like correspondence, so A1 is weak. The exact internal zeta has no target-divisor test and no ordinary Koopman Fredholm owner, so A2 fails. Target functional equation, Gamma factor, counting law, and continuation are absent, so A3 fails. The natural unitary action supplies only an A4 formal hint. Thus

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(A1_WEAK, A2_FAIL, A3_FAIL, A4_FORMAL_HINT),
ROUTE_A_EXPLORATORY,  route_b_invocation_allowed=false.
```

We make no claim about arithmetic/local data, Euler factors, root numbers, automorphy, Hilbert–Pólya, Riemann zeros, natural quantization, or Route B. Scope: `NO_BAD_EULER_OR_ROOT_NUMBER`.